

PolarLink

Integrated Polar Expedition Logistics
and Asset Management System

Backend research, architecture, USPs and prototype plan

ORGANIZATION

Ministry of Earth Sciences (MoES)

DEPARTMENT

National Centre for Polar and Ocean Research (NCPOR)

CATEGORY / THEME

Software · Smart Automation

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1. Executive summary

India runs year-round research stations in Antarctica (Maitri and Bharati) and the Arctic (Himadri). Every season, NCPOR moves thousands of tonnes of cargo, fuel and people across a multi-leg chain: Goa → Mumbai port or Mopa airport → Cape Town → ice-class ship or DROMLAN aircraft → the stations. Air and ship access exists only in the short austral summer. The winter-over team then lives on what reached them for several months, with only a thin satellite link to the outside world.

PolarLink is a centralized, offline-first platform for expedition planning, cargo chain-of-custody, station inventory, personnel movement and emergency response. It is designed around the three realities of polar operations: **seasonal access windows, unreliable low-bandwidth links, and zero margin for error in emergencies.**

The one-line pitch

Other systems tell you what you have. PolarLink tells you **how long each station can survive, what breaks if the ship is late, and the fastest way out in an emergency**, even when the internet is down.

Headline differentiators

- **Days of Autonomy:** inventory shown as survival time per station, compared against the next resupply window.
- **Satellite-grade sync:** priority lanes and a compact SOS packet designed to fit in a single satellite short-burst message.
- **What-if season simulator:** Monte Carlo simulation of delays that produces a plan confidence score.
- **International-aware evacuation engine:** routes through neighbouring stations and the actual air and sea network.
- **Tamper-evident custody ledger:** hash-chained audit trail for government procurement and CAG audits.
- **Glove-friendly field UX** and a **Madrid Protocol waste ledger.**

2. Problem understanding & domain research

Problem statement

Develop a centralized digital platform for expedition planning, cargo tracking, inventory management, personnel movement and emergency response (PS 26062, MoES / NCPOR, Software, Smart Automation).

2.1 Stations

Station	Location	Key facts relevant to logistics
Maitri (Antarctica)	Schirmacher Oasis, Central Dronning Maud Land	Inland, about 100 km from the ice shelf where ships berth. Crew of about 25. Close to the DROMLAN air gateway.
Bharati (Antarctica)	Larsemann Hills, Ingrid Christensen Coast	Coastal (Quilty Bay). Supports about 47 crew. Reachable by air only via Maitri.
Himadri (Arctic)	Ny-Ålesund, Svalbard	Arctic research base. Separate, simpler logistics chain.
NCPOR HQ	Vasco da Gama, Goa	Planning, procurement, packing, personnel briefing and training.

About 3,000 km separate Maitri and Bharati, so the two stations effectively need separate logistics plans.

2.2 The supply chain

Leg	Mode	Typical duration / notes
NCPOR Goa → Mumbai port	Road, containers	Food, provisions, spares and scientific equipment leave in lots from October to November.
Mumbai → Cape Town	Shipping liner	Transit to the gateway hub.
Goa (Mopa) → Cape Town → Antarctica	Direct air cargo (IL-76)	First direct air cargo mission in Oct 2025: about 18 t of equipment, medicines and provisions.
Cape Town → Bharati	Chartered ice-class ship	10–12 days, depending on weather and sea ice.
Bharati ↔ Maitri	Ship	5–7 days.
Cape Town ↔ Maitri	IL-76 (DROMLAN)	About 6 hours. Dates shift with Antarctic weather.
Maitri ↔ Bharati	Basler / Twin Otter (12 seats)	10–12 hours with a mid-way refuelling stop. Expensive and needs advance planning.
Cape Town procurement	Local vendors	About 600 barrels of Jet A1 plus lubricants tendered each year, plus ship bunker fuel.

2.3 Operational constraints that shape the design

Constraint	Implication for the software
Air operations only about Nov–Feb; ships only in summer	Every plan is bound to a season window. Late cargo means a months-long shortage.
Winter-over crew fully isolated	Stock accuracy and consumption forecasting are life-critical, not just bookkeeping.
Low-bandwidth, intermittent satellite link	The system must work offline at stations and sync opportunistically, with priorities.

Constraint	Implication for the software
Multi-leg, multi-vendor, multi-country chain	Chain of custody is needed at every hand-off, with QR scans and audit trails.
Bharati reachable by air only via Maitri	Evacuation needs graph-based route planning, not a simple lookup.
Neighbouring foreign stations nearby	International cooperation is part of emergency response.
Madrid Protocol waste-removal obligations	Waste has to be tracked as reverse cargo.
Large public spend, formal tenders	Audit-ready records and procurement linkage are needed.

2.4 Personnel flow

Members report to NCPOR Goa in batches, take briefing sessions and firefighting training, then travel via Cape Town. Teams are split into **summer** and **winter-over** groups. Winter-over members are de-inducted on DROMLAN flights at the start of the next season. Medical clearance, training status and current location are therefore core data.

2.5 Gaps in current practice (hypothesis to validate)

- Planning, cargo lists and stock registers are likely spread across spreadsheets, email and phone or satellite calls.
- There is no single live view of where every consignment is along the chain.
- Stock-out risk for winter is judged manually, not forecast.
- Emergency options are worked out ad hoc under pressure.
- Validate these points with the NCPOR contact listed on the problem statement before the final round.

3. Solution overview: PolarLink

PolarLink has six functional modules on one shared data model. It runs as an HQ node in Goa plus offline station nodes.

Module	What it does
Expedition Planner	Creates expeditions (e.g. 46-ISEA), legs, season windows, manifests and a timeline view.
Cargo & Custody	Handles the container → package → item hierarchy, QR tagging, scan-based state changes and a hash-chained custody log.
Station Inventory	Tracks stock per station, expiry batches, consumption logs, low-stock alerts and Days of Autonomy .
Personnel Movement	Covers nomination → medical → training → travel → station → de-induction, plus a live roll-call.
Assets & Maintenance	Manages snowmobiles, generators, vehicles and instruments, including condition and maintenance schedules.
Emergency Response	Handles SOS, incident lifecycle, muster roll-call, medical-stock lookup and the evacuation engine .

Smart Automation layer (cross-cutting)

- Consumption forecasting and the Days of Autonomy computation
- What-if season simulator (Monte Carlo)
- Load optimizer for aircraft and cargo holds (OR-Tools)
- Delay anomaly detection on custody timelines
- Automated alerts: low stock, expiry, missed scans, maintenance due, weather risk

4. Unique selling propositions (USPs)

Most teams will build a generic inventory and tracking dashboard. These USPs are aimed at the problems that are specific to polar operations.

USP 1 Days of Autonomy: a survival clock for every station

“Other systems tell you what you have. Ours tells you how long you’ll survive.”

- Converts stock into days remaining per critical resource: *Diesel 142 d · Food 210 d · Insulin 38 d (critical)*.
- Formula: current stock ÷ forecast daily consumption, adjusted for live headcount (summer vs. winter crew).
- Compares each value against the **next feasible resupply date** from the transport graph. Anything that runs out first turns red.
- Gives the director one number per station, and gives planners the reasoning behind it.

Prototype build: Low effort. Forecast endpoint + inventory + next open leg → one dashboard widget.

USP 2 Built for satellite links, not Wi-Fi

“Engineered for a 2.4 kbps link at minus 40, not for campus Wi-Fi.”

- **Priority lanes:** SOS and medical events sync first, then inventory and custody, and last photos and bulk logs.
- **Compact SOS packet:** station, type, severity, person ID, GPS and timestamp encoded in under 100 bytes, small enough for a single Iridium Short Burst Data message (capped at a few hundred bytes).
- **Delta sync:** only change events travel, compressed and batched.

Prototype build: Two FastAPI nodes + a bandwidth-throttle toggle. The SOS arrives immediately while bulk data queues.

USP 3 What-if season simulator

“Plan for the season you’ll actually get, not the one you hope for.”

- Planner asks: *what if the ship is delayed 12 days by sea ice at Bharati?*
- The system replays the plan and shows missed windows, stations that run short, and broken crew rotations.
- Monte Carlo over leg-delay distributions produces a **plan confidence score**, e.g. *82% chance all critical cargo arrives on time*.
- Turns the platform from record-keeping into decision support, which is what the Smart Automation theme asks for.

Prototype build: Medium effort. A single-scenario delay replay is enough for the demo; show the full Monte Carlo in slides.

USP 4 International-aware evacuation engine

“In Antarctica, your nearest help might fly a different flag.”

- The transport network is modelled as a graph. Edges carry duration, asset, capacity, season flag and weather gate.
- It includes neighbouring stations as nodes: Novolazarevskaya (Russia, next to Maitri, near the DROMLAN Novo runway), and Progress (Russia) and Zhongshan (China) near Bharati in the Larsemann Hills.
- On an SOS, it ranks routes by time and feasibility and matches the patient's needs against medicine stock, the doctor on site and the nearest better facility.
- Generates a one-click assistance request for the neighbouring station, in line with COMNAP-style cooperation.

Prototype build: Medium effort. networkx graph with 2–3 foreign nodes plus the Open-Meteo weather gate. Verify station facts before presenting.

USP 5 Tamper-evident chain of custody

“Every hand-off, provable. Audit-ready by design.”

- Each custody event stores the SHA-256 of the previous event. Editing history breaks the chain and is detected.
- One-click audit report per consignment: who, where, when, condition, photos.
- Deliberately **not** blockchain. You get the integrity benefit without the buzzword or the overhead.

Prototype build: Low effort. A few lines in the scan handler plus a verify endpoint.

USP 6 Glove-friendly field UX

“Designed for the person on the ice, not the person at the desk.”

- Offline PWA with large tap targets and scan-first flows: scanning a QR opens the next action directly.
- **Voice updates:** “Consumed 4 barrels diesel, generator 2” is parsed into a transaction and confirmed with one tap.
- High-contrast snow-glare mode.

Prototype build: Optional. The browser speech API plus a small rule-based parser is enough.

USP 7 Madrid Protocol waste ledger

“Leave nothing behind, and prove it.”

- Tracks waste as reverse cargo: generated → segregated → packed → loaded → returned.
- Auto-generates a seasonal environmental compliance summary by waste category.

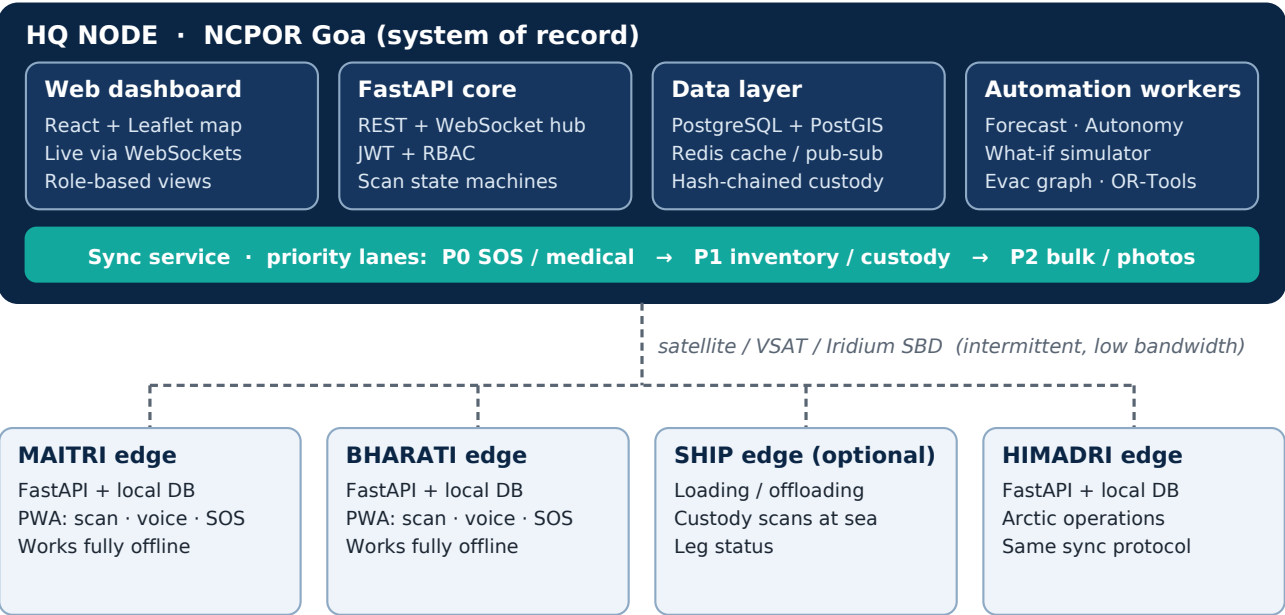
Prototype build: Slides only, or a simple extra consignment category.

USP build matrix

USP	Build in prototype	Show in slides	Judge impact
Days of Autonomy	Fully	-	Very high
Satellite-mode sync + compact SOS	Throttle demo	Iridium integration	Very high
What-if simulator	Single delay scenario	Full Monte Carlo	High
International evac engine	Graph with 2-3 foreign nodes	Assistance protocol	Very high
Hash-chained custody	Fully	-	Medium-high
Glove-friendly / voice	If time permits	UX mockups	Medium
Waste ledger	-	Yes	Medium

5. System architecture

PolarLink uses a **hub-and-edge** architecture. The HQ node is the system of record and runs analytics. Each station, and optionally the ship, runs an edge node that keeps working with no connectivity and syncs through an event log.



Each edge node keeps an append-only event log and syncs with HQ whenever the link is available.

Key design decisions

Decision	Why
Offline-first edge nodes	Stations must keep working through link outages and polar weather.
Event log instead of state replication	Easy to prioritise, compress, replay and audit. Avoids overwrite conflicts.
Inventory stored as deltas	Concurrent edits from two nodes merge correctly (a CRDT-like counter).
PostGIS	Station, route and geofence queries. Emergency location handling.

Decision	Why
Python backend	ML, optimisation and graph libraries live in the same codebase as the API.
WebSockets	Live SOS, roll-call and custody updates on the HQ dashboard.

6. Technology stack

Layer	Choice	Notes
API	FastAPI + Pydantic (async)	Auto OpenAPI docs, which also serve as a demo asset.
Database	PostgreSQL 16 + PostGIS	Edge nodes can use Postgres or SQLite for the prototype.
ORM / migrations	SQLAlchemy 2 + Alembic	
Cache / pub-sub	Redis	WebSocket fan-out and rate limiting.
Background jobs	Celery or ARQ	Sync, forecasts, simulations, alerts.
Forecasting	statsmodels (Holt-Winters) / Prophet	Per-station, per-item consumption.
Optimisation	Google OR-Tools	Aircraft and hold load planning.
Graph routing	networkx	Evacuation and resupply paths.
Weather	Open-Meteo API	Forecast at station coordinates, used as a route gate.
Auth	JWT + RBAC	Role and station scoped.
Frontend	React + Tailwind + Leaflet; PWA for field	Offline cache via service worker.
Deploy	Docker Compose	One command starts HQ plus two edge nodes.

7. Roles & access control

Role	Scope	Key permissions
HQ Logistics Officer	All	Plan expeditions and legs, create consignments, run simulations, view everything.
Expedition Leader	Expedition	Approve manifests, personnel assignments, incident command.
Station Leader	Own station	Station inventory, roll-call, assets, raise and manage incidents.
Store / Inventory Keeper	Own station	Receive scans, log consumption, stock counts.
Medical Officer	Own station	Medical inventory, personnel medical flags, medical incidents.
Ship / Air Ops Coordinator	Assigned legs	Load plans, leg status, scan loading and offloading.
Expedition Member	Self	Own itinerary, SOS, basic station info.

8. Data model

Entity	Key fields
Planning	
expeditions	id, code (e.g. 46-ISEA), type [antarctic arctic southern_ocean], season_start, season_end, status
stations	id, name, geom (PostGIS point), capacity, type [station ship hub port foreign_station], facilities_json
transport_assets	id, type [ship il76 basler twin_otter helicopter truck], capacity_kg, seats, status
legs	id, expedition_id, from_station, to_station, asset_id, planned/actual depart and arrive, season_only, status
Cargo	
consignments	id, qr_code, expedition_id, origin, destination, category [food fuel scientific medical spares waste], weight_kg, volume_m3, hazmat_class, temp_req, priority, status
packages	id, consignment_id, qr_code, contents_json, weight_kg
custody_events	id, consignment_id, leg_id, station_id, event_type, from_state, to_state, scanned_by, ts, geom, photo_url, prev_hash , hash (tamper-evident chain)
Inventory	
items	id, sku, name, unit, category, shelf_life_days, critical
inventory	station_id, item_id, batch, qty, min_threshold, expiry_date
inventory_txn	id, station_id, item_id, delta , reason [received consumed transfer wastage count_adj], event_uuid, lamport_ts, origin_node
People	
personnel	id, name, role, team [summer winter], medical_clearance, training_json, blood_group, emergency_contact
personnel_movements	id, person_id, leg_id, status [scheduled boarded arrived evacuated]
personnel_location	person_id, station_id, updated_at
Assets & emergency	
assets / maintenance_logs	id, tag, type, station_id, condition, next_maintenance / asset_id, action, by, ts, notes
incidents	id, type [medical fire missing_person equipment weather], severity, station_id, geom, person_id, reported_by, status, created_at
incident_actions	id, incident_id, action, by, ts
Sync	
sync_log	event_uuid, origin_node, entity, op, payload, lamport_ts, priority, applied_at

9. State machines

9.1 Consignment lifecycle

PLANNED → PACKED@Goa → IN_TRANSIT_TO_PORT → AT_PORT (Mumbai / Mopa) → SHIPPED
→ AT_HUB (Cape Town) → LOADED (ship / aircraft) → OFFLOADED → RECEIVED@Station
→ CONSUMED | RETROGRADE | WASTE_RETURNED

Every transition is triggered by a QR scan and writes a hash-chained custody_event. Illegal jumps are rejected. A consignment that stays in one state longer than its expected duration is flagged as an anomaly.

9.2 Personnel lifecycle

NOMINATED → MEDICALLY_CLEARED → TRAINED (firefighting, survival) → REPORTED@Goa → IN_TRANSIT
→ AT_STATION → (WINTERING) → DE_INDUCTED ↘ EVACUATED (from any in-field state)

9.3 Incident lifecycle

RAISED → ACKNOWLEDGED → MUSTER_COMPLETE → RESPONSE_ACTIVE → (EVACUATING) → RESOLVED → REVIEWED

10. API surface (prototype)

Area	Endpoints
Auth	POST /auth/login · GET /auth/me
Expeditions	CRUD /expeditions · GET /expeditions/{id}/timeline
Legs	CRUD /legs · GET /legs/{id}/manifest (people + cargo)
Cargo	POST /consignments · POST /scan/{qr} · GET /consignments/{id}/custody · GET /consignments/{id}/verify
Inventory	GET /stations/{id}/inventory · POST /inventory/txn · GET /stations/{id}/alerts
Autonomy	GET /stations/{id}/autonomy (days remaining per critical item vs. next resupply)
Personnel	CRUD /personnel · POST /movements · GET /stations/{id}/rollcall
Assets	CRUD /assets · POST /assets/{id}/maintenance
Emergency	POST /sos · POST /sos/compact (binary) · GET /incidents/{id}/evac-options · WS /ws/incidents
Simulation / ML	GET /forecast/{station}/{item} · POST /simulate/season · POST /optimize/load · GET /weather/{station}
Sync	POST /sync/push · GET /sync/pull?since=<lamport>&priority=<p> · GET /sync/status

11. AI / ML components

Component	Method	Input → Output	Priority
Consumption forecast	Holt-Winters / Prophet, headcount as a regressor	Consumption history + crew plan → daily forecast per item	P0
Days of Autonomy	Stock ÷ forecast, compared with next open leg	Inventory + forecast + graph → days left + risk flag	P0
What-if simulator	Discrete-event replay + Monte Carlo on leg delays	Plan + delay distributions → failures + confidence %	P1
Evacuation engine	Constrained shortest path (networkx) with weather and season gates	SOS + graph + weather → ranked routes	P1
Load optimizer	Multiple knapsack / bin packing (OR-Tools CP-SAT)	Consignments + asset limits → load plan	P1
Delay anomaly	Robust z-score / Isolation Forest on dwell times	Custody events → flagged consignments	P2
Voice / NL entry	Speech API + rule-based parser (optional LLM)	Utterance → inventory transaction	P2

Days of Autonomy: core logic

```
def autonomy(station, item):
    stock = sum(b.qty for b in inventory(station, item) if not expired(b))
    daily_use = forecast(station, item, horizon=180,
                        headcount=crew_plan(station)) # daily series
    days_left = first_day_where(cumsum(daily_use) >= stock)
    next_drop = earliest_arrival(graph, to=station, carrying=item,
                                after=today()) # season-aware
    if days_left < next_drop.days: risk = "CRITICAL"
    elif days_left < next_drop.days * 1.25: risk = "WATCH"
    else: risk = "OK"
    return {"item": item, "days_left": days_left,
            "next_resupply_in": next_drop.days, "risk": risk}
```

Evacuation ranking score

```
score(route) = w1·total_time + w2·(1 - weather_ok_prob) + w3·transfers
               + w4·medical_gap(route.destination)
# feasible only if every edge is in season and its asset is available
```

12. Offline sync protocol

- Every write on a node creates an event: {uuid, origin_node, entity, op, payload, lamport_ts, priority}.
- Events are queued locally in three lanes: **P0** SOS and medical, **P1** inventory and custody, **P2** bulk data and photos.
- When the link is up, the node pushes in lane order within a byte budget, then pulls events newer than its last acknowledged Lamport timestamp.
- **Merge rules:** inventory deltas are summed (commutative). Custody events are appended (hash-verified). Entity fields use last-writer-wins by Lamport timestamp and node ID. Semantic conflicts, such as one asset assigned at two stations, go to a review queue.
- Idempotent by UUID, so replays are safe.

- **Compact SOS** uses a fixed binary layout: station (1 B) · type (1 B) · severity (1 B) · person (2 B) · lat/lon (8 B) · ts (4 B) · CRC (2 B), about 19 bytes.

13. Dataset strategy

The problem statement has no dataset link, so PolarLink ships with a **synthetic but realistic seed generator**. The schema is designed to take in real NCPOR records later.

- About 200 items across food, fuel, medical, spares and scientific categories, with shelf lives and criticality flags.
- Three seasons of daily consumption history per station, generated with seasonality, a summer/winter headcount effect and noise.
- About 60 personnel with roles, teams, medical and training status.
- Transport graph: Goa, Mopa, Mumbai, Cape Town, Maitri, Bharati, Himadri, a vessel, and 2–3 neighbouring foreign stations.
- Historical leg delay distributions, used by the simulator.
- **Action:** email the NCPOR contact on the PS to ask about current formats (cargo lists, stock registers).

14. Prototype scope & build plan

Priority	Features
P0: must demo	Auth + RBAC · expeditions and legs · consignments + QR scan + custody timeline · station inventory with alerts · Days of Autonomy · personnel roll-call · SOS with live WebSocket alert · forecast endpoint
P1: differentiators	Offline edge node + prioritised sync + throttle toggle · compact SOS · evacuation engine · single-scenario what-if · hash-chain verification · load optimizer
P2: stretch	Monte Carlo confidence score · asset maintenance · waste ledger · delay anomaly · voice entry

Build order

#	Milestone	Output
1	Scaffold	Docker Compose, FastAPI, Postgres + PostGIS, Alembic, auth, seed generator
2	Cargo & custody	Consignment CRUD, scan state machine, hash chain, custody timeline
3	Inventory & autonomy	Transactions, alerts, forecast job, Days of Autonomy endpoint + widget
4	Personnel & emergency	Movements, roll-call, SOS, WebSocket hub, incident lifecycle
5	Graph intelligence	Transport graph, evacuation ranking, weather gate, what-if replay
6	Edge sync	Second node, event log, lanes, throttle toggle, compact SOS
7	Polish	Dashboard, map, demo data story, slides, rehearsal

Suggested team split for 4–6 people: backend core (1–2), ML and simulation (1), frontend and PWA (1–2), pitch and research (1).

15. Demo script & pitch structure

Hook (20 seconds)

“It's June at Bharati. Minus 40, 24-hour darkness, no ship for five months. A crew member needs insulin that's running low. What does your system do?”

Step	What judges see	Time
1	HQ creates 46-ISEA, its legs and consignments, and QR tags are generated	0:30
2	A consignment is scanned Goa → Mumbai → Cape Town → ship while the live map and custody timeline update	0:30
3	Days of Autonomy at Maitri flags diesel and insulin as CRITICAL before winter	0:30
4	What-if: ship delayed 12 days → shortages appear and the plan confidence score drops	0:30
5	Cut the Maitri link, log consumption offline, reconnect, and the sync reconciles correctly	0:30
6	Bandwidth throttled to satellite mode: an SOS at Bharati still arrives instantly at HQ	0:20
7	Evacuation engine ranks routes, including the neighbouring-station option and the Bharati-via-Maitri air route	0:30
8	Tamper test: edit a custody record in the DB, and verification catches it	0:10

Pitch flow

- **Problem:** scattered records, invisible delays, ad-hoc emergencies, all under hard seasonal constraints.
- **Insight:** polar logistics is a race against the season window over a weak link.
- **Solution + live demo** (the script above).
- **Why deployable:** offline-first, low bandwidth, audit-ready, mapped to NCPOR's actual Goa → Cape Town → station workflow.
- **Scale:** Himadri, Southern Ocean cruises, the Himalayan station, and other national programmes.

16. Risks, pitfalls & judge Q&A

Avoid

- Blockchain, drones or IoT everywhere with no clear need.
- A generic chatbot presented as the AI feature.
- 3D globes and visual effects that don't add function.
- Claiming real-time GPS on ship cargo mid-ocean. Tracking is scan-based at hand-off points.

Likely judge questions

Question	Answer outline
Where does your data come from?	Synthetic seed now. The schema maps to cargo lists, stock registers and manifests. Import adapters for CSV/Excel.
What if two stations edit the same record offline?	Deltas for stock, append-only custody, Lamport last-writer-wins for fields, review queue for semantic conflicts.

Question	Answer outline
How accurate is the forecast?	Backtested on held-out seasons. Shown with an interval. Autonomy uses a conservative percentile.
Is this secure?	JWT + RBAC scoped by station, TLS, hash-chained custody, audit logs. No SOS data is dropped.
Why not just use an existing ERP?	ERPs assume constant connectivity and don't model season windows, polar transport graphs or evacuation.
How would it connect to real satellite links?	The sync layer is transport-agnostic: HTTPS over VSAT today, compact SOS over Iridium SBD as a fallback.

17. Future scope

- Iridium SBD / VSAT gateway integration and hardware SOS beacons.
- Sea-ice and satellite-imagery feeds to predict ship delays more accurately.
- RFID and weight sensors at station stores for automatic stock counts.
- Integration with GeM / e-procurement for tender-to-delivery traceability.
- Shared situational awareness with neighbouring stations through COMNAP-aligned data exchange.
- Crew wellbeing and fatigue tracking for winter-over teams (with consent).
- Carbon and fuel-efficiency analytics per season.

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- [4] WION, coverage of the IL-76 cargo mission and station capacities (Maitri about 25, Bharati 47).
- [5] FESCO / SeaNews, NCPOR ship charter delivering more than 4,000 t of cargo and removing waste (2019).
- [6] Protocol on Environmental Protection to the Antarctic Treaty (Madrid Protocol), Annex III: waste disposal and management.
- [7] COMNAP, Council of Managers of National Antarctic Programs: operational cooperation.
- [8] Iridium Short Burst Data service documentation: message size limits.

Note: facts about neighbouring foreign stations and satellite message limits should be re-verified from official sources before the final presentation.